Lab 15 - Fuzzy Skin - Textured Walls on a Vase

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5 - apply fuzzy skin as a deliberate surface-texture treatment

Goal

Enable fuzzy skin on a vase model so the outer wall is jittered into a stone-like texture that hides layer lines, and print it with buildplate adhesion.

What you'll build

A vase with a uniform fuzzy texture that masks layer lines completely.

Tools: Ultimaker Cura, open vase.stl, 1.5mm-wall-vase.stl, FDM printer

Steps

1. Open 'open vase.stl' in Cura.

```
File > Open > open vase.stl
```

2. Enable fuzzy skin - the slicer adds random jitter to the outer wall toolpath.

```
Experimental > Fuzzy Skin ✔
```

3. Set fuzzy skin thickness to 0.3 mm and density to 1.25/mm for a fine stone texture.

```
Fuzzy Skin Thickness 0.3 mm | Density 1.25
```

4. Add a brim so the tall thin vase stays anchored.

```
Build Plate Adhesion > Brim
```

5. Slice and inspect the jittered outer wall in Preview.

```
Slice > Preview > outer wall
```

6. Print the vase and examine the texture - layer lines should be invisible within the fuzz.

```
print + inspect
```

7. Open 1.5mm-wall-vase.stl and verify the chosen fuzzy-skin thickness will not consume the thin wall; reduce the effect if Preview shows unstable or missing wall paths.

```
Open 1.5mm-wall-vase.stl > Preview > Line Type > inspect wall continuity
```

Evidence to capture

- Cura Preview showing fuzzy-skin jitter and the 1.5 mm wall toolpaths.
- Close-up photo and tactile assessment of the finished texture.

Test it

The printed vase carries an even tactile texture, and the 1.5 mm diagnostic vase retains continuous walls in Preview-showing that the texture setting fits the available wall thickness.

Files in this folder

- open vase.stl - Core sample model used by the lab procedure.
- 1.5mm-wall-vase.stl - Thin-walled vase benchmark for comparing fuzzy skin with wall-thickness constraints.